

Capstone Proposal: Measuring What Drives Citation in AI Search Engines

Mehreen Ali Gillani

Supervised by: Professor George Hagstrom

Section 1 — Introduction

AI search engines like ChatGPT, Perplexity, and Claude now answer questions directly instead of returning a list of links: they retrieve web pages, read them, and generate a summary that cites only a few sources. This has created a new practice, Generative Engine Optimization (GEO) (Aggarwal et al., 2024), aimed at making content more likely to be cited by these engines. GEO is far harder to measure than traditional SEO: there is no equivalent of Google Search Console; the same prompt produces different answers each time it is run, and different engines cite different sources for the same query. A single measurement tells you almost nothing about a page's true visibility. This matters to two audiences: publishers and marketers already investing in GEO with no reliable way to measure it, and researchers studying whether LLM-mediated search introduces new forms of instability or bias into how information reaches users.

Recent work has begun to map this problem. Schulte et al. (2026) found only 34–42% overlap in cited sources between repeated runs of the same prompt. C-SEO Bench (Puerto et al., 2025) found that most content-level GEO techniques are ineffective, and that the dominant factor is a page's initial ranking in traditional search. A recent survey (Martinez, 2026) concludes that no technique reliably makes content discoverable in the first place; optimization only helps after a page is already retrieved. My own preliminary data supports this: across 40 prompts run three times each through Perplexity's Pro Search; only 14% of retrieved URLs were stable across runs.

This project asks two questions:

- (1) How stable is AI retrieval and citation across repeated runs of the same prompt?
- (2) Does adding or removing one source change the relative probability that two other sources get cited?

Section 2 — Data

I have already collected three datasets and will extend them during the project.

AI citation data. Queried three engines: OpenAI (GPT-5.6 Luna), Perplexity (Sonar Pro), and Claude (Haiku 4.5) across 100 prompts spanning health, finance, travel, and software topics, producing 1,728 citations across 898 unique domains (fields: prompt, engine, cited URL, domain).

Google comparison data. Top-10 Google results per prompt via SerpAPI (765 pages), serving as a non-cited comparison group, pages that rank on Google but may or may not be cited by AI.

Controlled document-queue experiments. I will build a controlled experiment dataset in which I manually specify which documents each model sees, using chatlas or ellmer.

Section 3 — Data Analysis Plan

Processing. Clean all three datasets, strip tracking parameters from URLs, and match cited against retrieved URLs per prompt and run; remove pages under 100 words.

Exploratory analysis. Compare engines by citation count, domain type, retrieval and citation overlap; visualize retrieval and citations stability by prompt and topic.

Test 1 — Retrieval and Citation Stability Across Repeated Runs

For each prompt, I will run the query 5–8 times per engine (Schulte et al., 2026) across Perplexity, Claude, and OpenAI, computing Jaccard similarity and Rank-Biased Overlap (RBO) on retrieved and cited URL sets where each is exposed. This tests whether retrieval and citation are repeatable, and whether the two differ in stability. A methodological constraint shapes this test: OpenAI exposes only final citations; Claude exposes both retrieval and citation distinctly; Perplexity's two signals require careful extraction to distinguish, and an early check suggests they may overlap heavily, a result I will validate before treating it as a finding.

Test 2 — Independence of Irrelevant Alternatives (IIA)

Using a controlled document queue, I will show each AI search engine a fixed document set, then add or remove one document, and measure whether the relative citation odds of the remaining documents change; they should not, under IIA. Document order will be randomized across repeats to control for position sensitivity (Kumar & Lakkaraju, 2024). I will fit a Plackett-Luce model (Benson, Kumar & Tomkins, 2016) to estimate each document quality parameter per composition, and test for IIA violations by comparing relative worth ratios across compositions using a chi-square specification test. A small single-pass pilot (one 5-document set) suggested a violation, but this needs repetition across multiple compositions before being treated as a real result.

Test 3 — Predicting Citation from Rank and Content Features

I will fit a logistic regression predicting whether a page is cited, using rank and a small set of content features (word count, presence of statistics, and text perplexity computed with a pretrained language model, motivated by evidence that retrievers favor low-perplexity text; Wang et al., 2025). The model will be fit first on the controlled documents, where I set the feature values myself, and then on the Google comparison dataset (top 10 SerpAPI results for each prompt, matched against the citations collected from the three engines for the same prompts) to check whether the same features predict citation in naturally occurring pages. Comparing model fit with and without the content features shows how much they add beyond rank alone. If time allows, I will add a gradient boosting model as a robustness check. Because real pages differ in many ways, I do not measure; the Google-based results are correlational and will be reported as such.

Expected Contribution

This project contributes: (1) a reproducible measurement framework for AI search visibility across multiple deployed engines, accounting for the stochastic noise inherent in LLM-generated answers; (2) empirical evidence on how much of an engine's citation behavior is shaped by content-level features versus retrieval rank and randomness alone; and (3) an empirical test of whether LLM citation behavior satisfies the Independence of Irrelevant Alternatives, or is context-dependent on the competing document set.

Section 4 — Timeline and Risks

Over 12 weeks: weeks 1–2 complete data collection; weeks 3–4 run Test 1; weeks 5–7 run Test 2; weeks 8–9 run Test 3; weeks 10–12 write up results. By the midterm point (week 6), Test 1 should be complete, and Test 2 underway; success at the final stage means a documented, statistically supported answer to both research questions. The main risk is scope: three engines $\times$ three tests are ambitious for one semester if time is tight, I will complete IIA testing fully on two engines rather than three. A secondary risk is API changes or rate limits mid-project, mitigated by collecting incrementally and saving raw responses.

References

- Aggarwal, P., et al. (2024). GEO: Generative Engine Optimization. arXiv:2311.09735.
- Benson, A. R., Kumar, R., & Tomkins, A. (2016). On the Relevance of Irrelevant Alternatives. WWW 2016.
- Kumar, A., & Lakkaraju, H. (2024). Manipulating Large Language Models to Increase Product Visibility. arXiv:2404.07981.
- Martinez, A. (2026). Optimizing Visibility in Generative Engines: A Critical Survey of Generative Engine Optimization. arXiv:2607.14035.
- Puerto, H., et al. (2025). C-SEO Bench: Does Conversational SEO Work? arXiv:2506.11097.
- Schulte, J., Bleeker, M., & Kaufmann, P. (2026). Don't Measure Once: Measuring Visibility in AI Search (GEO). arXiv:2604.07585.
- Wang, H., et al. (2025). Perplexity Trap: PLM-Based Retrievers Overrate Low Perplexity Documents. ICLR 2025.
- Wu, Y., Zhong, S., Kim, Y., & Xiong, C. (2025). AutoGEO: Learning GEO Policies with Large Language Models. arXiv:2510.11438.
- Chen, Z., et al. (2026). Statistical Analysis of Website Features Impacting Retrieval Chances. arXiv:2604.27790.
- Pfrommer, S., Bai, Y., Gautam, T., & Sojoudi, S. (2024). Ranking Manipulation for Conversational Search Engines. EMNLP 2024.